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Section 11

CSE340

Assignment 4

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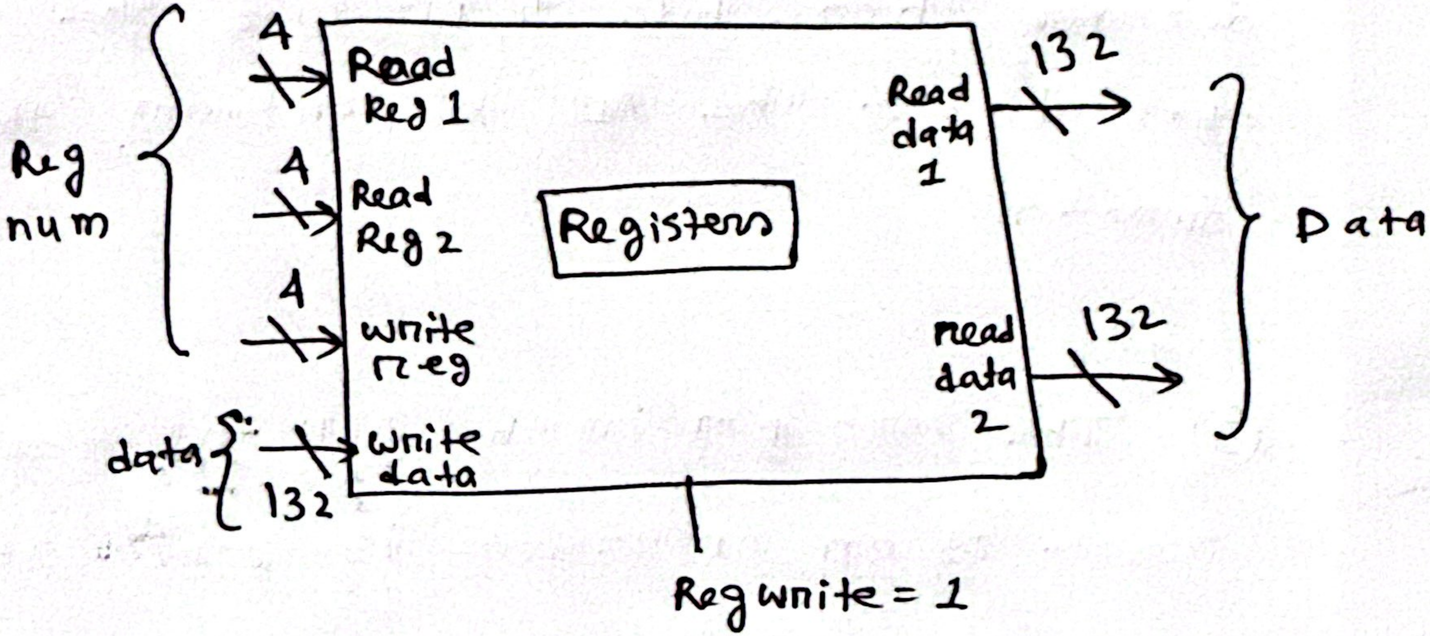
Reg num
4
Read Reg 1
4
Read Reg 2
4
write reg
data
132
write data
Registers
Read data 1
132
read data 2
132
Data
Regwrite = 1

(2) The ALU control unit determines the specific operation (like add, subtract, AND, OR) that the Arithmetic Logic Unit (ALU) will perform. It receives inputs from the instruction decoder to know what operation to perform. The bits Instruction[30, 14-12] are part of the instruction's op-code and function fields. The specific bits from the instruction vary depending on the instruction set architecture (ISA). In many architectures, these bits are crucial because they directly specify the desired ALU operation for that instruction. The ALU control unit decodes these bits to produce the control signals that direct the ALU to perform the correct function.

(3) The zero pin in the ALU indicates if the result of an operation is zero. It outputs 1 when the result is zero, otherwise 0. This is essential for branching, comparisons and controlling program flow.

(4) For single cycle:

cycle time = IF + RR + ALU + Mem + RW

= 20 + 20 + 40 + 40 + 20 = 140 sec

For pipeline:

cycle time = max (20, 20, 40, 20, 40)

= 40 sec

Pipeline will run the instruction faster.

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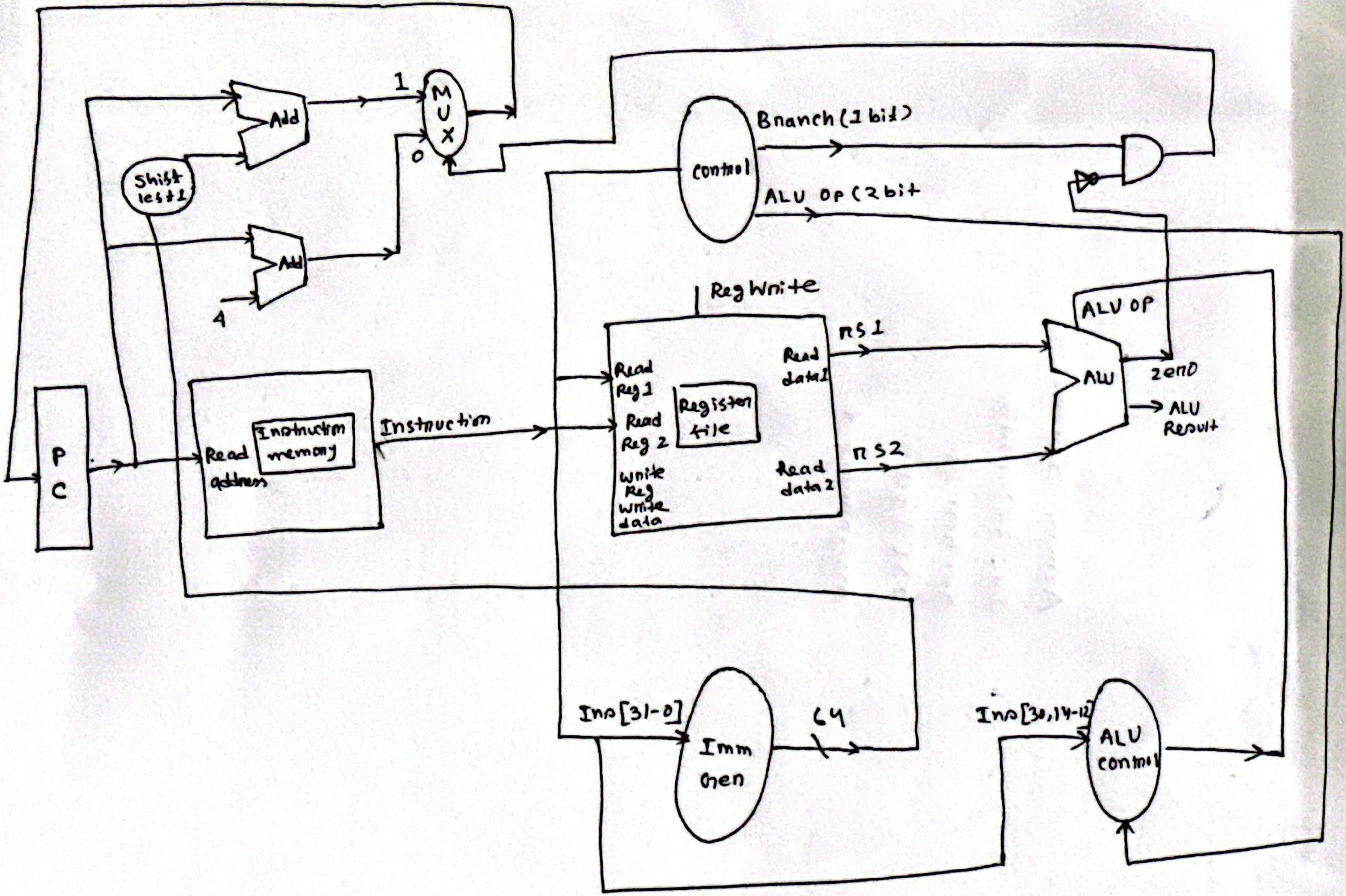

1
0
M
U
X
Add
Add
Shift left 1
4
P
C
Read address
Instruction memory
Instruction
control
Branch (1 bit)
ALU op (2 bit
RegWrite
Read Reg 1
Read Reg 2
Write Reg
Write data
Register file
Read data1
Read data2
rs1
rs2
ALU
ALU OP
zero
ALU Result
Ins[31-0]
Imm Gen
64
Ins[30,14-12]
ALU control